

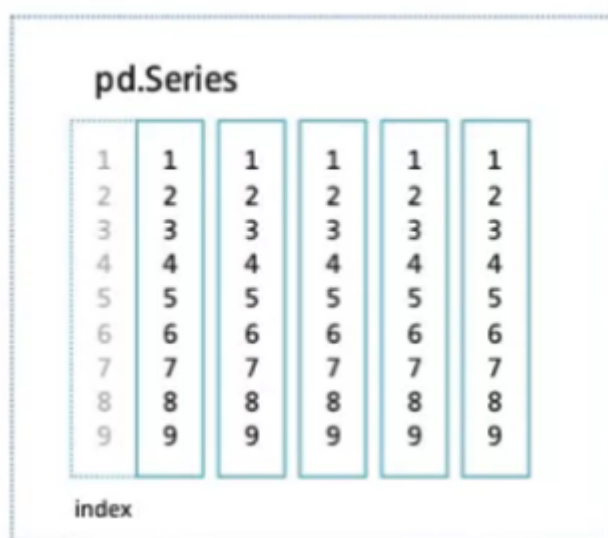
Machine Learning in Finance

Week- 2

Python Data Analysis Library

Pandas is an open source library providing high-performance, easy-to-use structures and data analysis tools for the Python programming language

pd.DataFrame



```
# For Google Colab only!
```

```
!cat /proc/driver/nvidia/gpus/0000:00:04.0/information
```

In []:

In []: #

```
Model:          Unknown
IRQ:            11
GPU UUID:       GPU-9698ba26-4fdc-6ed0-8b14-41a4ab8a6034
Video BIOS:     90.04.a7.00.01
Bus Type:       PCI
DMA Size:       47 bits
DMA Mask:       0x7fffffffffff
Bus Location:   0000:00:04.0
Device Minor:   0
GPU Excluded:   No
```

```

# Use code in this cell for Google Colab only

# Run the cell and upload the dataset
# adult_data_mini.csv

from google.colab import files
uploaded = files.upload()
.....

```

```
In [ ]: from google.colab import files
        uploaded = files.upload()
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun cell to enable.

Saving adult_data_mini-1.csv to adult_data_mini-1.csv

```
In [ ]: #
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun cell to enable.

Saving adult_data_mini.csv to adult_data_mini (1).csv

Data Frame

```

# Check the uploaded dataset
ls
.....

```

```
In [ ]: ls
```

adult_data_mini-1.csv [sample_data/](#)

```
In [ ]: #
```

'adult_data_mini (1).csv' adult_data_mini.csv [sample_data/](#)

```

import pandas as pd
.....

```

```
In [ ]: import pandas as pd
```

```
In [ ]: #
```

```

# create a DataFrame

df = pd.DataFrame({'numbers':range(8), 'chars':['g']*8})
.....

```

```
In [ ]: df = pd.DataFrame({'numbers':range(8), 'chars':['g']*8})
```

```
In [ ]: #
```

```
# print the DataFrame
```

```
.....df
```

```
In [ ]: df
```

```
Out[ ]:
```

	numbers	chars
0	0	g
1	1	g
2	2	g
3	3	g
4	4	g
5	5	g
6	6	g
7	7	g

```
In [ ]: #
```

```
Out[ ]:
```

	numbers	chars
0	0	g
1	1	g
2	2	g
3	3	g
4	4	g
5	5	g
6	6	g
7	7	g

```
# clearing the memory occupied by the DataFrame
```

```
df = []  
df  
.....
```

```
In [ ]: df = []  
df
```

```
Out[ ]: []
```

```
In [ ]: #
```

Out[]: []

.....

Creating and printing a DataFrame by reading data from a *.csv file

```
data = pd.read_csv('adult_data_mini.csv', header=0)
data
```

```
In [ ]: data = pd.read_csv('adult_data_mini-1.csv', header=0)
data
```

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife
5	37	Private	284582	Masters	14	Married-civ-spouse	Exec-managerial	Wife
6	49	Private	160187	9th	5	Married-spouse-absent	Other-service	Not-in-family
7	52	Self-emp-not-inc	209642	HS-grad	9	Married-civ-spouse	Exec-managerial	Husband
8	31	Private	45781	Masters	14	Never-married	Prof-specialty	Not-in-family
9	42	Private	159449	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband
10	37	Private	280464	Some-college	10	Married-civ-spouse	Exec-managerial	Husband
11	30	State-gov	141297	Bachelors	13	Married-civ-spouse	Prof-specialty	Husband
12	23	Private	122272	Bachelors	13	Never-married	Adm-clerical	Own-child
13	32	Private	205019	Assoc-acdm	12	Never-married	Sales	Not-in-family
14	40	Private	121772	Assoc-voc	11	Married-civ-spouse	Craft-repair	Husband

In []: #

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife
5	37	Private	284582	Masters	14	Married-civ-spouse	Exec-managerial	Wife
6	49	Private	160187	9th	5	Married-spouse-absent	Other-service	Not-in-family
7	52	Self-emp-not-inc	209642	HS-grad	9	Married-civ-spouse	Exec-managerial	Husband
8	31	Private	45781	Masters	14	Never-married	Prof-specialty	Not-in-family
9	42	Private	159449	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband
10	37	Private	280464	Some-college	10	Married-civ-spouse	Exec-managerial	Husband
11	30	State-gov	141297	Bachelors	13	Married-civ-spouse	Prof-specialty	Husband
12	23	Private	122272	Bachelors	13	Never-married	Adm-clerical	Own-child
13	32	Private	205019	Assoc-acdm	12	Never-married	Sales	Not-in-family
14	40	Private	121772	Assoc-voc	11	Married-civ-spouse	Craft-repair	Husband

```
# size of the DataFrame
```

```
.....data.shape
```

```
In [ ]: data.shape
```

```
Out[ ]: (15, 15)
```

```
In [ ]: #
```

```
Out[ ]: (15, 15)
```

```
# Quick view for the DataFrame structure (first 5 row
```

```
.....data.head()
```

```
In [ ]: data.head()
```

```
Out[ ]:
```

	age	workclass	fnlwgt	education	education- num	marital- status	occupation	relationship	
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	W
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	W
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	W
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	B
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	B

```
In [ ]: #
```

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship	sex
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	W
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	W
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	W
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	B
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	B

In []:

DataFrame Modification

.....

Create a new row as a dictionary

```
New_Line = {'age': '25', 'workclass': 'Private', 'education': 'Some-college', 'race': 'White', 'sex': 'Male'}
print(New_Line)
print(type(New_Line))
```

In []:

```
New_Line = {'age': '25', 'workclass': 'Private', 'education': 'Some-college', 'sex': 'Male'}
print(New_Line)
print(type(New_Line))
```

```
{'age': '25', 'workclass': 'Private', 'education': 'Some-college', 'race': 'W
'sex': 'Male'}
<class 'dict'>
```

In []:

```
#
{'age': '25', 'workclass': 'Private', 'education': 'Some-college', 'race': 'W
'sex': 'Male'}
<class 'dict'>
```

.....

transform the dictionary into a pandas-Series and then into a pandas DataF

```
New_Line_series = pd.Series(New_Line)
print(type(New_Line_series))
print(New_Line_series, '\n')
```

```
New_Line_dataframe = New_Line_series.to_frame().T
print(type(New_Line_dataframe))
New_Line_dataframe
```

```
In [ ]: New_Line_series = pd.Series(New_Line)
print(type(New_Line_series))
print(New_Line_series, '\n')

New_Line_dataframe = New_Line_series.to_frame().T
print(type(New_Line_dataframe))
New_Line_dataframe
```

```
<class 'pandas.core.series.Series'>
age                25
workclass          Private
education          Some-college
race               White
sex               Male
dtype: object
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Out[ ]:      age  workclass  education  race  sex
0    25    Private  Some-college  White  Male
```

```
In [ ]: #
```

```
<class 'pandas.core.series.Series'>
age                25
workclass          Private
education          Some-college
race               White
sex               Male
dtype: object
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Out[ ]:      age  workclass  education  race  sex
0    25    Private  Some-college  White  Male
```

.....

Add the New_Line_dataframe to the end of the 'data' DataFrame

```
data = pd.concat([data, New_Line_dataframe], ignore_index=True)
data
```

```
In [ ]: data = pd.concat([data, New_Line_dataframe], ignore_index=True)
data
```

Out[]:

	age	workclass					fnlwgt	education	education-num	marital-status	occupation	relationsh
0	39	State-gov	7	7	5	1	6.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in fam
1	50	Self-emp-not-inc	8	3	3	1	1.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husband
2	38	Private	2	1	5	6	4.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in fam
3	53	Private	2	3	4	7	2.0	11th	7.0	Married-civ-spouse	Handlers-cleaners	Husband
4	28	Private	3	3	8	4	0.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	Wife
5	37	Private	2	8	4	5	8.0	Masters	14.0	Married-civ-spouse	Exec-managerial	Wife
6	49	Private	1	6	0	1	8.0	9th	5.0	Married-spouse-absent	Other-service	Not-in fam
7	52	Self-emp-not-inc	2	0	9	6	4.0	HS-grad	9.0	Married-civ-spouse	Exec-managerial	Husband
8	31	Private	4	5	7	8	1.0	Masters	14.0	Never-married	Prof-specialty	Not-in fam
9	42	Private	1	5	9	4	4.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husband
10	37	Private	2	8	0	4	6.0	Some-college	10.0	Married-civ-spouse	Exec-managerial	Husband
11	30	State-gov	1	4	1	2	9.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	Husband
12	23	Private	1	2	2	2	7.0	Bachelors	13.0	Never-married	Adm-clerical	Own-child
13	32	Private	2	0	5	0	1.0	Assoc-acdm	12.0	Never-married	Sales	Not-in fam
14	40	Private	1	2	1	7	7.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husband
15	25	Private	NaN				NaN		NaN	NaN	NaN	NaN

age	workclass	fnlwgt	education	education- num	marital- status	occupation	relationsl
			Some- college				

In []: #

Out[]:

	age	workclass					fnlwgt	education	education-num	marital-status	occupation	relationsh
0	39	State-gov	7	7	5	1	6.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in fam
1	50	Self-emp-not-inc	8	3	3	1	1.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husba
2	38	Private	2	1	5	6	4.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in fam
3	53	Private	2	3	4	7	2.0	11th	7.0	Married-civ-spouse	Handlers-cleaners	Husba
4	28	Private	3	3	8	4	0.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	W
5	37	Private	2	8	4	5	8.0	Masters	14.0	Married-civ-spouse	Exec-managerial	W
6	49	Private	1	6	0	1	8.0	9th	5.0	Married-spouse-absent	Other-service	Not-in fam
7	52	Self-emp-not-inc	2	0	9	6	4.0	HS-grad	9.0	Married-civ-spouse	Exec-managerial	Husba
8	31	Private	4	5	7	8	1.0	Masters	14.0	Never-married	Prof-specialty	Not-in fam
9	42	Private	1	5	9	4	4.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husba
10	37	Private	2	8	0	4	6.0	Some-college	10.0	Married-civ-spouse	Exec-managerial	Husba
11	30	State-gov	1	4	1	2	9.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	Husba
12	23	Private	1	2	2	2	7.0	Bachelors	13.0	Never-married	Adm-clerical	Own-ch
13	32	Private	2	0	5	0	1.0	Assoc-acdm	12.0	Never-married	Sales	Not-in fam
14	40	Private	1	2	1	7	7.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husba
15	25	Private	NaN				NaN				NaN	NaN

age	workclass	fnlwgt	education	education- num	marital- status	occupation	relationsl
			Some- college				

```
In [ ]: # If you use the default value
        # ignore_index=False,
        # then the row index at the end will be -0-

        # Delete the 10th and 12th rows using drop-method
        # For rows use Axis=0, for columns - Axis=1
        # To save changings, use inplace=True

        data.drop([9,11], axis=0, inplace=True)
        data
        .....
```

```
In [ ]: data.drop([9,11], axis=0, inplace=True)
        data
```

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
0	39	State-gov	77516.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in fam
1	50	Self-emp-not-inc	83311.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husba
2	38	Private	215646.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in fam
3	53	Private	234721.0	11th	7.0	Married-civ-spouse	Handlers-cleaners	Husba
4	28	Private	338409.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	W
5	37	Private	284582.0	Masters	14.0	Married-civ-spouse	Exec-managerial	W
6	49	Private	160187.0	9th	5.0	Married-spouse-absent	Other-service	Not-in fam
7	52	Self-emp-not-inc	209642.0	HS-grad	9.0	Married-civ-spouse	Exec-managerial	Husba
8	31	Private	45781.0	Masters	14.0	Never-married	Prof-specialty	Not-in fam
10	37	Private	280464.0	Some-college	10.0	Married-civ-spouse	Exec-managerial	Husba
12	23	Private	122272.0	Bachelors	13.0	Never-married	Adm-clerical	Own-ch
13	32	Private	205019.0	Assoc-acdm	12.0	Never-married	Sales	Not-in fam
14	40	Private	121772.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husba
15	25	Private	NaN	Some-college	NaN	NaN	NaN	NaN

In []: #

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
0	39	State-gov	77516.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in fam
1	50	Self-emp-not-inc	83311.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husband
2	38	Private	215646.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in fam
3	53	Private	234721.0	11th	7.0	Married-civ-spouse	Handlers-cleaners	Husband
4	28	Private	338409.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	Wife
5	37	Private	284582.0	Masters	14.0	Married-civ-spouse	Exec-managerial	Wife
6	49	Private	160187.0	9th	5.0	Married-spouse-absent	Other-service	Not-in fam
7	52	Self-emp-not-inc	209642.0	HS-grad	9.0	Married-civ-spouse	Exec-managerial	Husband
8	31	Private	45781.0	Masters	14.0	Never-married	Prof-specialty	Not-in fam
10	37	Private	280464.0	Some-college	10.0	Married-civ-spouse	Exec-managerial	Husband
12	23	Private	122272.0	Bachelors	13.0	Never-married	Adm-clerical	Own-child
13	32	Private	205019.0	Assoc-acdm	12.0	Never-married	Sales	Not-in fam
14	40	Private	121772.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husband
15	25	Private	NaN	Some-college	NaN	NaN	NaN	NaN

```
# Add a new column

data['IsHomeless'] = [False]*13 + [True]
data
.....
```

```
In [ ]: data['IsHomeless'] = [False]*13 + [True]
data
```

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
0	39	State-gov	77516.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in fam
1	50	Self-emp-not-inc	83311.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husband
2	38	Private	215646.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in fam
3	53	Private	234721.0	11th	7.0	Married-civ-spouse	Handlers-cleaners	Husband
4	28	Private	338409.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	Wife
5	37	Private	284582.0	Masters	14.0	Married-civ-spouse	Exec-managerial	Wife
6	49	Private	160187.0	9th	5.0	Married-spouse-absent	Other-service	Not-in fam
7	52	Self-emp-not-inc	209642.0	HS-grad	9.0	Married-civ-spouse	Exec-managerial	Husband
8	31	Private	45781.0	Masters	14.0	Never-married	Prof-specialty	Not-in fam
10	37	Private	280464.0	Some-college	10.0	Married-civ-spouse	Exec-managerial	Husband
12	23	Private	122272.0	Bachelors	13.0	Never-married	Adm-clerical	Own-child
13	32	Private	205019.0	Assoc-acdm	12.0	Never-married	Sales	Not-in fam
14	40	Private	121772.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husband
15	25	Private	NaN	Some-college	NaN	NaN	NaN	NaN

In []: #

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
0	39	State-gov	77516.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in fam
1	50	Self-emp-not-inc	83311.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husband
2	38	Private	215646.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in fam
3	53	Private	234721.0	11th	7.0	Married-civ-spouse	Handlers-cleaners	Husband
4	28	Private	338409.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	Wife
5	37	Private	284582.0	Masters	14.0	Married-civ-spouse	Exec-managerial	Wife
6	49	Private	160187.0	9th	5.0	Married-spouse-absent	Other-service	Not-in fam
7	52	Self-emp-not-inc	209642.0	HS-grad	9.0	Married-civ-spouse	Exec-managerial	Husband
8	31	Private	45781.0	Masters	14.0	Never-married	Prof-specialty	Not-in fam
10	37	Private	280464.0	Some-college	10.0	Married-civ-spouse	Exec-managerial	Husband
12	23	Private	122272.0	Bachelors	13.0	Never-married	Adm-clerical	Own-child
13	32	Private	205019.0	Assoc-acdm	12.0	Never-married	Sales	Not-in fam
14	40	Private	121772.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husband
15	25	Private	NaN	Some-college	NaN	NaN	NaN	NaN

check the new size

..... data.shape

```
In [ ]: data.shape
```

```
Out[ ]: (14, 16)
```

```
In [ ]: #
```

```
Out[ ]: (14, 16)
```

```
# print the column information
```

```
..... data.dtypes
```

```
In [ ]: data.dtypes
```

```
Out[ ]:
```

	0
age	object
workclass	object
fnlwgt	float 6 4
education	object
education-num	float 6 4
marital-status	object
occupation	object
relationship	object
race	object
sex	object
capital-gain	float 6 4
capital-loss	float 6 4
hours-per-week	float 6 4
native-country	object
Answer	object
IsHomeless	bool

```
dtype: object
```

```
In [ ]: #
```

Out[]: 0

age	object
workclass	object
fnlwgt	float 6 4
education	object
education-num	float 6 4
marital-status	object
occupation	object
relationship	object
race	object
sex	object
capital-gain	float 6 4
capital-loss	float 6 4
hours-per-week	float 6 4
native-country	object
Answer	object
IsHomeless	bool

dtype: object

Print more of the DataFrame information:

..... data.info()

In []: data.info()

```

<class 'pandas.core.frame.DataFrame'>
Index: 14 entries, 0 to 15
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                    14 non-null    object
1   workclass              14 non-null    object
2   fnlwgt                 13 non-null    float64
3   education              14 non-null    object
4   education-num          13 non-null    float64
5   marital-status         13 non-null    object
6   occupation             13 non-null    object
7   relationship           13 non-null    object
8   race                   14 non-null    object
9   sex                    14 non-null    object
10  capital-gain            4 non-null     float64
11  capital-loss            3 non-null     float64
12  hours-per-week          13 non-null    float64
13  native-country          13 non-null    object
14  Answer                  13 non-null    object
15  IsHomeless              14 non-null    bool
dtypes: bool(1), float64(5), object(10)
memory usage: 1.8+ KB

```

In []: #

```

<class 'pandas.core.frame.DataFrame'>
Index: 14 entries, 0 to 15
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                    14 non-null    object
1   workclass              14 non-null    object
2   fnlwgt                 13 non-null    float64
3   education              14 non-null    object
4   education-num          13 non-null    float64
5   marital-status         13 non-null    object
6   occupation             13 non-null    object
7   relationship           13 non-null    object
8   race                   14 non-null    object
9   sex                    14 non-null    object
10  capital-gain            4 non-null     float64
11  capital-loss            3 non-null     float64
12  hours-per-week          13 non-null    float64
13  native-country          13 non-null    object
14  Answer                  13 non-null    object
15  IsHomeless              14 non-null    bool
dtypes: bool(1), float64(5), object(10)
memory usage: 1.8+ KB

```

Missing Data

```
# NaN (Not a Number) - it's missing data.  
# It's highly inconvenient to analyse missing data.  
# Fill the 'capital-gain' column with zeros.  
  
data['capital-gain'] = data['capital-gain'].fillna(0)  
data
```

.....

```
In [ ]: data['capital-gain'] = data['capital-gain'].fillna(0)  
data
```

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
0	39	State-gov	77516.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in fam
1	50	Self-emp-not-inc	83311.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husba
2	38	Private	215646.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in fam
3	53	Private	234721.0	11th	7.0	Married-civ-spouse	Handlers-cleaners	Husba
4	28	Private	338409.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	W
5	37	Private	284582.0	Masters	14.0	Married-civ-spouse	Exec-managerial	W
6	49	Private	160187.0	9th	5.0	Married-spouse-absent	Other-service	Not-in fam
7	52	Self-emp-not-inc	209642.0	HS-grad	9.0	Married-civ-spouse	Exec-managerial	Husba
8	31	Private	45781.0	Masters	14.0	Never-married	Prof-specialty	Not-in fam
10	37	Private	280464.0	Some-college	10.0	Married-civ-spouse	Exec-managerial	Husba
12	23	Private	122272.0	Bachelors	13.0	Never-married	Adm-clerical	Own-ch
13	32	Private	205019.0	Assoc-acdm	12.0	Never-married	Sales	Not-in fam
14	40	Private	121772.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husba
15	25	Private	NaN	Some-college	NaN	NaN	NaN	NaN

In []: #

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
0	39	State-gov	77516.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in fam
1	50	Self-emp-not-inc	83311.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husband
2	38	Private	215646.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in fam
3	53	Private	234721.0	11th	7.0	Married-civ-spouse	Handlers-cleaners	Husband
4	28	Private	338409.0	Bachelors	13.0	Married-civ-spouse	Prof-specialty	Wife
5	37	Private	284582.0	Masters	14.0	Married-civ-spouse	Exec-managerial	Wife
6	49	Private	160187.0	9th	5.0	Married-spouse-absent	Other-service	Not-in fam
7	52	Self-emp-not-inc	209642.0	HS-grad	9.0	Married-civ-spouse	Exec-managerial	Husband
8	31	Private	45781.0	Masters	14.0	Never-married	Prof-specialty	Not-in fam
10	37	Private	280464.0	Some-college	10.0	Married-civ-spouse	Exec-managerial	Husband
12	23	Private	122272.0	Bachelors	13.0	Never-married	Adm-clerical	Own-child
13	32	Private	205019.0	Assoc-acdm	12.0	Never-married	Sales	Not-in fam
14	40	Private	121772.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husband
15	25	Private	NaN	Some-college	NaN	NaN	NaN	NaN

Data slice manipulating

```
# Print just one column
```

```
.....data.education
```

```
In [ ]: data.education
```

```
Out[ ]:
```

	education
0	Bachelors
1	Bachelors
2	HS-grad
3	1 1 th
4	Bachelors
5	Masters
6	9 th
7	HS-grad
8	Masters
1 0	Some-college
1 2	Bachelors
1 3	Assoc-acdm
1 4	Assoc-voc
1 5	Some-college

```
dtype: object
```

```
In [ ]: #
```

Out[]:

education	
0	Bachelors
1	Bachelors
2	HS-grad
3	1 1 th
4	Bachelors
5	Masters
6	9 th
7	HS-grad
8	Masters
1 0	Some-college
1 2	Bachelors
1 3	Assoc-acdm
1 4	Assoc-voc
1 5	Some-college

dtype: object

```
# Another way to print one column
```

```
data['capital-gain']
```

.....

In []: data['capital-gain']

Out[]: capital-gain

0	2 1 7 4 . 0
1	0 . 0
2	0 . 0
3	0 . 0
4	0 . 0
5	0 . 0
6	0 . 0
7	0 . 0
8	1 4 0 8 4 . 0
1 0	0 . 0
1 2	0 . 0
1 3	0 . 0
1 4	0 . 0
1 5	0 . 0

dtype: float 6 4

In []: #

Out[]:

		capital-gain
0	2 1 7 4 . 0	
1		0 . 0
2		0 . 0
3		0 . 0
4		0 . 0
5		0 . 0
6		0 . 0
7		0 . 0
8	1 4 0 8 4 . 0	
1 0		0 . 0
1 2		0 . 0
1 3		0 . 0
1 4		0 . 0
1 5		0 . 0

dtype: float 6 4

Or...

.....data[['capital-gain']]

In []: data[['capital-gain']]

Out[]: capital-gain

0	2 1 7 4 . 0
1	0 . 0
2	0 . 0
3	0 . 0
4	0 . 0
5	0 . 0
6	0 . 0
7	0 . 0
8	1 4 0 8 4 . 0
1 0	0 . 0
1 2	0 . 0
1 3	0 . 0
1 4	0 . 0
1 5	0 . 0

In []: #

Out[]:

	capital-gain
0	2174.0
1	0.0
2	0.0
3	0.0
4	0.0
5	0.0
6	0.0
7	0.0
8	14084.0
10	0.0
12	0.0
13	0.0
14	0.0
15	0.0

```
# Demonstrate two or more columns
```

```
data[['occupation', 'capital-gain']]
```

In []: `data[['occupation', 'capital-gain']]`

Out[]:

	occupation	capital-gain
0	Adm-clerical	2 1 7 4 . 0
1	Exec-managerial	0 . 0
2	Handlers-cleaners	0 . 0
3	Handlers-cleaners	0 . 0
4	Prof-specialty	0 . 0
5	Exec-managerial	0 . 0
6	Other-service	0 . 0
7	Exec-managerial	0 . 0
8	Prof-specialty	1 4 0 8 4 . 0
1 0	Exec-managerial	0 . 0
1 2	Adm-clerical	0 . 0
1 3	Sales	0 . 0
1 4	Craft-repair	0 . 0
1 5	NaN	0 . 0

In []: #

```
Out[ ]:
```

	occupation	capital-gain
0	Adm-clerical	2 1 7 4 . 0
1	Exec-managerial	0 . 0
2	Handlers-cleaners	0 . 0
3	Handlers-cleaners	0 . 0
4	Prof-specialty	0 . 0
5	Exec-managerial	0 . 0
6	Other-service	0 . 0
7	Exec-managerial	0 . 0
8	Prof-specialty	1 4 0 8 4 . 0
1 0	Exec-managerial	0 . 0
1 2	Adm-clerical	0 . 0
1 3	Sales	0 . 0
1 4	Craft-repair	0 . 0
1 5	NaN	0 . 0

```
# Take the first three rows
```

```
data[:3]
```

```
In [ ]: data[:3]
```

```
Out[ ]:
```

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship
0	3 9	State-gov	7 7 5 1 6 . 0	Bachelors	1 3 . 0	Never-married	Adm-clerical	Not-in-family
1	5 0	Self-emp-not-inc	8 3 3 1 1 . 0	Bachelors	1 3 . 0	Married-civ-spouse	Exec-managerial	Husband
2	3 8	Private	2 1 5 6 4 6 . 0	HS-grad	9 . 0	Divorced	Handlers-cleaners	Not-in-family

```
In [ ]: #
```

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship
0	39	State-gov	77516.0	Bachelors	13.0	Never-married	Adm-clerical	Not-in-family
1	50	Self-emp-not-inc	83311.0	Bachelors	13.0	Married-civ-spouse	Exec-managerial	Husband
2	38	Private	215646.0	HS-grad	9.0	Divorced	Handlers-cleaners	Not-in-family

Take the last three rows

data[-3:]

.....

In []: data[-3:]

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship
13	32	Private	205019.0	Assoc-acdm	12.0	Never-married	Sales	Not-in-family
14	40	Private	121772.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husband
15	25	Private	NaN	Some-college	NaN	NaN	NaN	Not-in-family

In []: #

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship
13	32	Private	205019.0	Assoc-acdm	12.0	Never-married	Sales	Not-in-family
14	40	Private	121772.0	Assoc-voc	11.0	Married-civ-spouse	Craft-repair	Husband
15	25	Private	NaN	Some-college	NaN	NaN	NaN	Not-in-family

Another way to take the last three rows

data.tail(3)

.....

```
In [ ]: data.tail(3)
```

```
Out[ ]:
```

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh	
1	3	3 2	Private	2 0 5 0 1 9 . 0	Assoc-acdm	1 2 . 0	Never-married	Sales	Not-i fam
1	4	4 0	Private	1 2 1 7 7 2 . 0	Assoc-voc	1 1 . 0	Married-civ-spouse	Craft-repair	Husba
1	5	2 5	Private		NaN	Some-college	NaN	NaN	NaN

```
In [ ]: #
```

```
Out[ ]:
```

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh	
1	3	3 2	Private	2 0 5 0 1 9 . 0	Assoc-acdm	1 2 . 0	Never-married	Sales	Not-i fam
1	4	4 0	Private	1 2 1 7 7 2 . 0	Assoc-voc	1 1 . 0	Married-civ-spouse	Craft-repair	Husba
1	5	2 5	Private		NaN	Some-college	NaN	NaN	NaN

Take the specified rows and columns

```
..... data.loc[[0,1,2], ["education", "occupation"]]
```

```
In [ ]: data.loc[[0,1,2], ["education", "occupation"]]
```

```
Out[ ]:
```

	education	occupation
0	Bachelors	Adm-clerical
1	Bachelors	Exec-managerial
2	HS-grad	Handlers-cleaners

```
In [ ]: #
```

```
Out[ ]:
```

	education	occupation
0	Bachelors	Adm-clerical
1	Bachelors	Exec-managerial
2	HS-grad	Handlers-cleaners

```
# Or...
```

```
.....data.iloc[[1,2,5], [0,3]]
```

```
In [ ]: data.iloc[[1,2,5], [0,3]]
```

```
Out[ ]:
```

	age	education
1	5 0	Bachelors
2	3 8	HS-grad
5	3 7	Masters

```
In [ ]: #
```

```
Out[ ]:
```

	age	education
1	5 0	Bachelors
2	3 8	HS-grad
5	3 7	Masters

```
# Take the specified element
```

```
.....data['education'].loc[2]
```

```
In [ ]: data['education'].loc[2]
```

```
Out[ ]: ' HS-grad'
```

```
In [ ]: #
```

```
Out[ ]: ' HS-grad'
```

```
# Select and print the Row
```

```
.....data.iloc[7]
```

```
In [ ]: data.iloc[7]
```

Out[]: 7

age	52
workclass	Self-emp-not-inc
fnlwgt	209642.0
education	HS-grad
education-num	9.0
marital-status	Married-civ-spouse
occupation	Exec-managerial
relationship	Husband
race	White
sex	Male
capital-gain	0.0
capital-loss	0.0
hours-per-week	45.0
native-country	United-States
Answer	> 50K
IsHomeless	False

dtype: object

In []: #

```
Out[ ]: 7
```

age	52
workclass	Self-emp-not-inc
fnlwgt	209642.0
education	HS-grad
education-num	9.0
marital-status	Married-civ-spouse
occupation	Exec-managerial
relationship	Husband
race	White
sex	Male
capital-gain	0.0
capital-loss	0.0
hours-per-week	45.0
native-country	United-States
Answer	> 50K
IsHomeless	False

dtype: object

Selection with conditions

```
.....
# Select and print the rows, which are equal to the condition (more or equal t
data[data['hours-per-week'] >= 50]
```

```
In [ ]: data[data['hours-per-week'] >= 50]
```

Out[]:

age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
8 3 1	Private	4 5 7 8 1 . 0	Masters	1 4 . 0	Never-married	Prof-specialty	Not-in fam
1 0 3 7	Private	2 8 0 4 6 4 . 0	Some-college	1 0 . 0	Married-civ-spouse	Exec-managerial	Husband
1 3 3 2	Private	2 0 5 0 1 9 . 0	Assoc-acdm	1 2 . 0	Never-married	Sales	Not-in fam

In []: #

Out[]:

age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
8 3 1	Private	4 5 7 8 1 . 0	Masters	1 4 . 0	Never-married	Prof-specialty	Not-in fam
1 0 3 7	Private	2 8 0 4 6 4 . 0	Some-college	1 0 . 0	Married-civ-spouse	Exec-managerial	Husband
1 3 3 2	Private	2 0 5 0 1 9 . 0	Assoc-acdm	1 2 . 0	Never-married	Sales	Not-in fam

.....

Select rows, which correspond to several conditions:

```
data[(data['hours-per-week'] >= 50) & (data['education-num'] > 11)]
```

In []: data[(data['hours-per-week'] >= 50) & (data['education-num'] > 11)]

Out[]:

age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationshi
8 3 1	Private	4 5 7 8 1 . 0	Masters	1 4 . 0	Never-married	Prof-specialty	Not-in fam
1 3 3 2	Private	2 0 5 0 1 9 . 0	Assoc-acdm	1 2 . 0	Never-married	Sales	Not-in fam

In []: #

Out[]:

	age	workclass		fnlwgt	education	education-num	marital-status	occupation	relationship
	8	3 1	Private	4 5 7 8 1.0	Masters	1 4.0	Never-married	Prof-specialty	Not-in-family
	1 3	3 2	Private	2 0 5 0 1 9.0	Assoc-acdm	1 2.0	Never-married	Sales	Not-in-family

.....

Select rows combining several conditions (logical OR)

data[(data['hours-per-week'] >= 50) | (data.IsHomeless == True)]

In []: data[(data['hours-per-week'] >= 50) | (data.IsHomeless == True)]

Out[]:

	age	workclass		fnlwgt	education	education-num	marital-status	occupation	relationship
	8	3 1	Private	4 5 7 8 1.0	Masters	1 4.0	Never-married	Prof-specialty	Not-in-family
	1 0	3 7	Private	2 8 0 4 6 4.0	Some-college	1 0.0	Married-civ-spouse	Exec-managerial	Husband
	1 3	3 2	Private	2 0 5 0 1 9.0	Assoc-acdm	1 2.0	Never-married	Sales	Not-in-family
	1 5	2 5	Private		NaN	NaN	NaN	NaN	NaN

In []: #

Out[]:

	age	workclass		fnlwgt	education	education-num	marital-status	occupation	relationship
	8	3 1	Private	4 5 7 8 1.0	Masters	1 4.0	Never-married	Prof-specialty	Not-in-family
	1 0	3 7	Private	2 8 0 4 6 4.0	Some-college	1 0.0	Married-civ-spouse	Exec-managerial	Husband
	1 3	3 2	Private	2 0 5 0 1 9.0	Assoc-acdm	1 2.0	Never-married	Sales	Not-in-family
	1 5	2 5	Private		NaN	NaN	NaN	NaN	NaN

```
# Select a slice of the initial DataFrame
```

```
data.loc[[7,8],['education','occupation']]
```

```
In [ ]: data.loc[[7,8],['education','occupation']]
```

```
Out[ ]:
```

	education	occupation
7	HS-grad	Exec-managerial
8	Masters	Prof-specialty

```
In [ ]: #
```

```
Out[ ]:
```

	education	occupation
7	HS-grad	Exec-managerial
8	Masters	Prof-specialty

```
# Print unique values in the column
```

```
data['native-country'].unique()
```

```
In [ ]: data['native-country'].unique()
```

```
Out[ ]: array([' United-States', ' Cuba', ' Jamaica', ' ?', nan], dtype=object)
```

```
In [ ]: #
```

```
Out[ ]: array([' United-States', ' Cuba', ' Jamaica', ' ?', nan], dtype=object)
```

```
.....
```

```
# Function count() selects the number of each unique value
```

```
data['native-country'].value_counts()
```

```
In [ ]: data['native-country'].value_counts()
```

Out[]: count

native-country	
United-States	1 0
Cuba	1
Jamaica	1
?	1

dtype: int64

In []: #

Out[]: count

native-country	
United-States	1 0
Cuba	1
Jamaica	1
?	1

dtype: int64

.....

number of characters in each row of the 'education' column

data['education'].apply(len)

In []: data['education'].apply(len)

Out[]: education

0	1 0
1	1 0
2	8
3	5
4	1 0
5	8
6	4
7	8
8	8
1 0	1 3
1 2	1 0
1 3	1 1
1 4	1 0
1 5	1 2

dtype: int 6 4

In []: #

Out[]:

education		
0	1	0
1	1	0
2		8
3		5
4	1	0
5		8
6		4
7		8
8		8
1 0	1	3
1 2	1	0
1 3	1	1
1 4	1	0
1 5	1	2

dtype: int 6 4

In []: *# Analyse the result above!*

Write your ideas about why more characters are counted in some rows

In []:

Sort by a column

inplace=False by default

ascending=True by default

.....`data.sort_values(by='hours-per-week', ascending=False`

In []: `data.sort_values(by='hours-per-week', ascending=False)`

Out[]:

	age	workclass		fnlwgt	education	education-num	marital-status	occupation	relationsh
	1 0	3 7	Private	2 8 0 4 6 4 . 0	Some-college	1 0 . 0	Married-civ-spouse	Exec-managerial	Husba
	8	3 1	Private	4 5 7 8 1 . 0	Masters	1 4 . 0	Never-married	Prof-specialty	Not-i fam
	1 3	3 2	Private	2 0 5 0 1 9 . 0	Assoc-acdm	1 2 . 0	Never-married	Sales	Not-i fam
	7	5 2	Self-emp-not-inc	2 0 9 6 4 2 . 0	HS-grad	9 . 0	Married-civ-spouse	Exec-managerial	Husba
	0	3 9	State-gov	7 7 5 1 6 . 0	Bachelors	1 3 . 0	Never-married	Adm-clerical	Not-i fam
	1 4	4 0	Private	1 2 1 7 7 2 . 0	Assoc-voc	1 1 . 0	Married-civ-spouse	Craft-repair	Husba
	4	2 8	Private	3 3 8 4 0 9 . 0	Bachelors	1 3 . 0	Married-civ-spouse	Prof-specialty	W
	3	5 3	Private	2 3 4 7 2 1 . 0	1 1 th	7 . 0	Married-civ-spouse	Handlers-cleaners	Husba
	2	3 8	Private	2 1 5 6 4 6 . 0	HS-grad	9 . 0	Divorced	Handlers-cleaners	Not-i fam
	5	3 7	Private	2 8 4 5 8 2 . 0	Masters	1 4 . 0	Married-civ-spouse	Exec-managerial	W
	1 2	2 3	Private	1 2 2 2 7 2 . 0	Bachelors	1 3 . 0	Never-married	Adm-clerical	Own-ch
	6	4 9	Private	1 6 0 1 8 7 . 0	9 th	5 . 0	Married-spouse-absent	Other-service	Not-i fam
	1	5 0	Self-emp-not-inc	8 3 3 1 1 . 0	Bachelors	1 3 . 0	Married-civ-spouse	Exec-managerial	Husba
	1 5	2 5	Private		NaN	Some-college	NaN	NaN	NaN

In []: #

Out[]:

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationsh
1 0	3 7	Private	2 8 0 4 6 4 . 0	Some-college	1 0 . 0	Married-civ-spouse	Exec-managerial	Husba
8	3 1	Private	4 5 7 8 1 . 0	Masters	1 4 . 0	Never-married	Prof-specialty	Not-i fam
1 3	3 2	Private	2 0 5 0 1 9 . 0	Assoc-acdm	1 2 . 0	Never-married	Sales	Not-i fam
7	5 2	Self-emp-not-inc	2 0 9 6 4 2 . 0	HS-grad	9 . 0	Married-civ-spouse	Exec-managerial	Husba
0	3 9	State-gov	7 7 5 1 6 . 0	Bachelors	1 3 . 0	Never-married	Adm-clerical	Not-i fam
1 4	4 0	Private	1 2 1 7 7 2 . 0	Assoc-voc	1 1 . 0	Married-civ-spouse	Craft-repair	Husba
4	2 8	Private	3 3 8 4 0 9 . 0	Bachelors	1 3 . 0	Married-civ-spouse	Prof-specialty	W
3	5 3	Private	2 3 4 7 2 1 . 0	1 1 th	7 . 0	Married-civ-spouse	Handlers-cleaners	Husba
2	3 8	Private	2 1 5 6 4 6 . 0	HS-grad	9 . 0	Divorced	Handlers-cleaners	Not-i fam
5	3 7	Private	2 8 4 5 8 2 . 0	Masters	1 4 . 0	Married-civ-spouse	Exec-managerial	W
1 2	2 3	Private	1 2 2 2 7 2 . 0	Bachelors	1 3 . 0	Never-married	Adm-clerical	Own-ch
6	4 9	Private	1 6 0 1 8 7 . 0	9 th	5 . 0	Married-spouse-absent	Other-service	Not-i fam
1	5 0	Self-emp-not-inc	8 3 3 1 1 . 0	Bachelors	1 3 . 0	Married-civ-spouse	Exec-managerial	Husba
1 5	2 5	Private		NaN	Some-college	NaN	NaN	NaN

Grouping

In []: `# groupby()` - the function groups the first column based on the second

.....

```
# Create a new DataFrame grouping data by sex and education
```

```
Group_by_education = data.groupby(["sex","education"])
Group_by_education.size()
```

```
In [ ]: Group_by_education = data.groupby(["sex","education"])
Group_by_education.size()
```

```
Out[ ]: 0
```

sex	education	
Female	9 th	1
	Bachelors	2
	Masters	2
Male	1 1 th	1
	Assoc-acdm	1
	Assoc-voc	1
	Bachelors	2
	HS-grad	2
	Some-college	1
Male	Some-college	1

dtype: int64

```
In [ ]: #
```

Out[]: 0

sex	education	
Female	9 th	1
	Bachelors	2
	Masters	2
Male	1 1 th	1
	Assoc-acdm	1
	Assoc-voc	1
	Bachelors	2
	HS-grad	2
	Some-college	1
Male	Some-college	1

dtype: int64

In []: *# As you see above, the string variables were not grouped correctly as th*

```
# Let's clear the 'sex' column from spaces:
```

```
data["sex"] = data["sex"].map(str.strip)
```

```
# And run the grouping one more time
```

```
Group_by_education = data.groupby(["sex", "education"])  
Group_by_education.size()
```

.....

In []: data["sex"] = data["sex"].map(str.strip)

```
Group_by_education = data.groupby(["sex", "education"])  
Group_by_education.size()
```

Out[]: 0

sex	education	
Female	9 th	1
	Bachelors	2
	Masters	2
Male	1 1 th	1
	Assoc-acdm	1
	Assoc-voc	1
	Bachelors	2
	HS-grad	2
	Some-college	1
	Some-college	1

dtype: int64

In []: #

In []: # The 'education' column also has data with spaces

```
# Let's clear the 'education' column from spaces:
```

```
data["education"] = data["education"].map(str.strip)
```

```
# And run the grouping one more time
```

```
Group_by_education = data.groupby(["sex", "education"])
Group_by_education.size()
```

.....

In []: data["education"] = data["education"].map(str.strip)

```
Group_by_education = data.groupby(["sex", "education"])
Group_by_education.size()
```

Out[]: 0

sex	education	
Female	9 th	1
	Bachelors	2
	Masters	2
Male	1 1 th	1
	Assoc-acdm	1
	Assoc-voc	1
	Bachelors	2
	HS-grad	2
	Some-college	2

dtype: int64

In []: #

Out[]: 0

sex	education	
Female	9 th	1
	Bachelors	2
	Masters	2
Male	1 1 th	1
	Assoc-acdm	1
	Assoc-voc	1
	Bachelors	2
	HS-grad	2
	Some-college	2

dtype: int64

In []: # Now it's ok

```

.....
# Create an asymmetric DataFrame grouping education and sex

Group_by_sex = data.groupby(['education', 'sex'])
print(type(Group_by_sex))
Group_by_sex.size()

```

```

In [ ]: Group_by_sex = data.groupby(['education', 'sex'])
        print(type(Group_by_sex))
        Group_by_sex.size()

```

```
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>
```

```
Out[ ]: 0
```

education	sex	
1 1 th	Male	1
9 th	Female	1
Assoc-acdm	Male	1
Assoc-voc	Male	1
Bachelors	Female	2
	Male	2
HS-grad	Male	2
Masters	Female	2
Some-college	Male	2

```
dtype: int64
```

```
In [ ]: #
```

```
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>
```

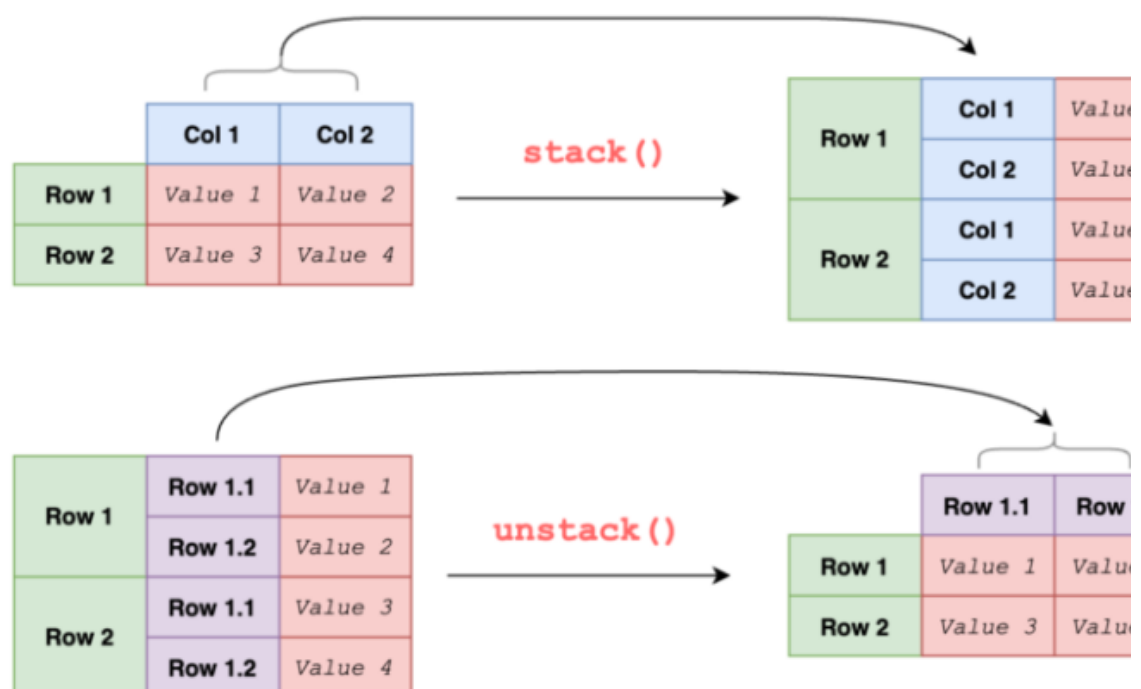
Out[]: 0

education	sex	
1 1 th	Male	1
9 th	Female	1
Assoc-acdm	Male	1
Assoc-voc	Male	1
Bachelors	Female	2
	Male	2
HS-grad	Male	2
Masters	Female	2
Some-college	Male	2

dtype: int64

```
In [ ]: # As you can see, Group_by_education is not a DataFrame, it has a special  
# To create a new DataFrame from DataFrameGroupBy, you need to apply the
```

- `stack()` : stack the prescribed level(s) from column to row.
- `unstack()` : unstack the prescribed level(s) from row to column. The inverse operation from stack.



.....

```
# Convert the DataFrameGroupBy format to a new DataFrame using unstack() method
```

```
Group_by_sex_DataFrame = Group_by_sex.size().unstack()  
Group_by_sex_DataFrame
```

```
In [ ]: Group_by_sex_DataFrame = Group_by_sex.size().unstack()  
Group_by_sex_DataFrame
```

```
Out[ ]:      sex  Female    Male
```

education			
1	1 th	NaN	1 . 0
9	th	1 . 0	NaN
Assoc-acdm		NaN	1 . 0
Assoc-voc		NaN	1 . 0
Bachelors	2 . 0	2 . 0	
HS-grad	NaN	2 . 0	
Masters	2 . 0	NaN	
Some-college	NaN	2 . 0	

```
In [ ]: #
```

```
Out[ ]:      sex  Female    Male
```

education			
1	1 th	NaN	1 . 0
9	th	1 . 0	NaN
Assoc-acdm		NaN	1 . 0
Assoc-voc		NaN	1 . 0
Bachelors	2 . 0	2 . 0	
HS-grad	NaN	2 . 0	
Masters	2 . 0	NaN	
Some-college	NaN	2 . 0	

Data Transformation

```
In [ ]: # The apply() method allows you to apply custom functions to data
```

```
df = pd.DataFrame({'A': [1, 2], 'B': [10, 20]})
df
.....
```

```
In [ ]: df = pd.DataFrame({'A' : [1,2], 'B' : [10,20]})
df
```

```
Out[ ]:
```

	A	B
0	1	10
1	2	20

```
In [ ]: #
```

```
Out[ ]:
```

	A	B
0	1	10
1	2	20

```
def custom_function(x):
    return x * 2

df['B'] = df['B'].apply(custom_function)
df
.....
```

```
In [ ]: def custom_function(x):
        return x*2

df['B'] = df['B'].apply(custom_function)
df
```

```
Out[ ]:
```

	A	B
0	1	20
1	2	40

```
In [ ]: #
```

```
Out[ ]:
```

	A	B
0	1	20
1	2	40

Pandas can read .csv or .html database from a webpage:

```
In [ ]: # This is a very useful feature for downloading financial and other data
```

.....

```
url= 'http://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/winequality-whi
data_2 = pd.read_csv(url, header=0, sep=';', low_memory=False) # The separator is ';'
data_2.head()
```

```
In [ ]: url= 'https://archive.ics.uci.edu/ml/machine-learning-databases/wine-qual
data_2 = pd.read_csv(url, header=0, sep=';', low_memory=False)
data_2.head()
```

```
Out[ ]:
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density
0	7.0	0.27	0.36	2.07	0.045	4.50	170.0	1.0010
1	6.3	0.30	0.34	1.6	0.049	14.0	132.0	0.9940
2	8.1	0.28	0.40	6.9	0.050	30.0	97.0	0.9951
3	7.2	0.23	0.32	8.5	0.058	47.0	186.0	0.9956
4	7.2	0.23	0.32	8.5	0.058	47.0	186.0	0.9956

```
In [ ]: #
```

```
Out[ ]:
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density
0	7.0	0.27	0.36	2.07	0.045	4.50	170.0	1.0010
1	6.3	0.30	0.34	1.6	0.049	14.0	132.0	0.9940
2	8.1	0.28	0.40	6.9	0.050	30.0	97.0	0.9951
3	7.2	0.23	0.32	8.5	0.058	47.0	186.0	0.9956
4	7.2	0.23	0.32	8.5	0.058	47.0	186.0	0.9956

```
In [ ]:
```

Lab Logbook Requirement:

- 1 . Determine a number (n) equal to the last digit of your SID. If the last digit of your SID is ' 0 ', then use 10 .
- 2 . Group by "relationship" and "hours-per-week". The first grouping should be by relationship.
- 3 . Reduce all "hours-per-week" column values in the original DataFrame by the value 'n'. Use a function.
- 4 . Group by "relationship" and the reduced "hours-per-week".
- 5 . Add the code and result to your Lab Logbook.

NOTE: DON'T FORGET TO SAVE AND BACK UP YOUR COMPLETE JUPYTER NOTEBOOK AND LAB LOGBOOK ON GITHUB.

```
In [ ]: # 1. Determine n from the last digit of the SID=2520731
n = 1

# 2. Group by relationship and hours-per-week
Group_by_relationship_hours = data.groupby(["relationship", "hours-per-week"])
Group_by_relationship_hours.size()

# 3. Reduce hours-per-week by n using a function
def reduce_hours(x):
    return x - n

data["hours-per-week-reduced"] = data["hours-per-week"].apply(reduce_hours)

data

# 4. Group by relationship and reduced hours-per-week
Group_by_relationship_reduced_hours = data.groupby(
    ["relationship", "hours-per-week-reduced"]
)

Group_by_relationship_reduced_hours.size()
```

Out[]: 0

relationship	hours-per-week-reduced		
Husband	1	2	0
	3	9	0
	4	4	0
	7	9	0
Not-in-family	1	5	0
	3	9	0
	4	9	0
Own-child	2	9	0
Wife	3	9	0

dtype: int 6 4

In []: